

# Clouds and fog

When air containing water vapor rises into the sky, it cools. Since cold air cannot carry as much water vapor as warm air, its water content condenses into a mass of tiny water droplets, which we see as a cloud. If the air at ground level is already saturated with water vapor, the vapor may condense to form low-level clouds known as fog.



## WATER VAPOR

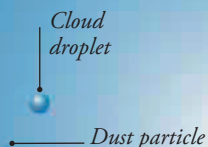
Water is always evaporating and rising into the air as an invisible gas, or vapor. Evaporation absorbs heat from the surrounding water, and if the vapor then turns back into liquid water, this heat is released. When water vapor condenses into the tiny droplets that form clouds, the heat is released into the air, warming it so that it rises and builds the clouds higher and higher.

## CLOUD DROPLETS

Cloud droplets condense around microscopic particles, such as dust or salt tossed into the air by ocean waves. The cloud droplets are so tiny that it takes a million of them to form one average raindrop. At high, cold altitudes, clouds are formed from microscopic ice crystals instead.

### ► DROPS AND DROPLETS

*The quarter circle represents part of a normal raindrop—a million times bigger than a cloud droplet.*



## MIST AND FOG

Clouds can develop at ground or sea level, filling the air with microscopic water droplets to form a veil of mist or fog, as seen here in San Francisco. This often happens when warm, moist air moves over a cold surface such as the sea. The air is cooled to a point at which some of the water vapor in it condenses.

